Latent macro-sensitivity creates "leaky" instruments

In practice, we rarely know the true factor loadings (λ_i)—how each firm responds to the macroeconomy.

- **The Risk:** If we assume $\lambda_i = 1$ (Simple GIV) when large firms are systematically more sensitive to macro shocks, the instrument becomes correlated with η_t.
- **The Consequence:** The exclusion restriction is violated, and IV estimates of elasticities are biased.
- **The Solution:** We must estimate the loadings ($\hat{\lambda}_i$) from the panel data to construct the orthogonal weights Γ.

PCA extracts factor loadings from firm-level outcomes

If we assume idiosyncratic shocks are small relative to common factors, we can use **Principal Component Analysis**.

1. **Panel Analysis:** Apply PCA to the $T \times N$ matrix of outcomes (L_{it}).
2. **Factor Extraction:** The first K principal components represent the latent macro shocks ($\hat{\eta}_t$).
3. **Loadings:** The coefficients of the firm outcomes on these components are the estimated loadings ($\hat{\lambda}_i$).
4. **Weighting:** Construct Γ by projecting firm sizes S orthogonal to $\hat{\lambda}$ and a constant.

Iterative GMM identifies parameters and loadings simultaneously

For a more robust approach, Gabaix and Koijen (2022) suggest joint estimation via the **Generalized Method of Moments**.

The Iterative Logic:

- **Step 1:** Start with an initial guess for the elasticity $\psi^{(0)}$.
- **Step 2:** Extract the "macro-contaminated" residuals from the demand equation.
- **Step 3:** Estimate λ_i from these residuals (e.g., via the first principal component).
- **Step 4:** Update the GIV instrument z_t using the new loadings and re-estimate $\psi^{(1)}$.

This process continues until the elasticity and factor loadings converge.

Choosing the estimation strategy in practice

Assumption	**Approach**	**Estimation Method**
Homogeneous Loadings	Simple GIV	2SLS (Difference-in-Averages)
Known Factor Loadings	Optimal GIV	2SLS with Analytical Γ
Unknown Loadings	**Latent GIV**	**PCA or Iterative GMM**

Practical Note: When using estimated loadings ($\hat{\lambda}$), standard errors should ideally be bootstrapped or adjusted to account for the "generated regressor" variance.